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Lee

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(54) **ELECTRONIC DEVICE COMPRISING
ELECTRONIC PEN STORAGE STRUCTURE,
AND ELECTRONIC PEN**

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(2013.01); **G06F 2200/1632** (2013.01)

(58) **Field of Classification Search**
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2200/1632

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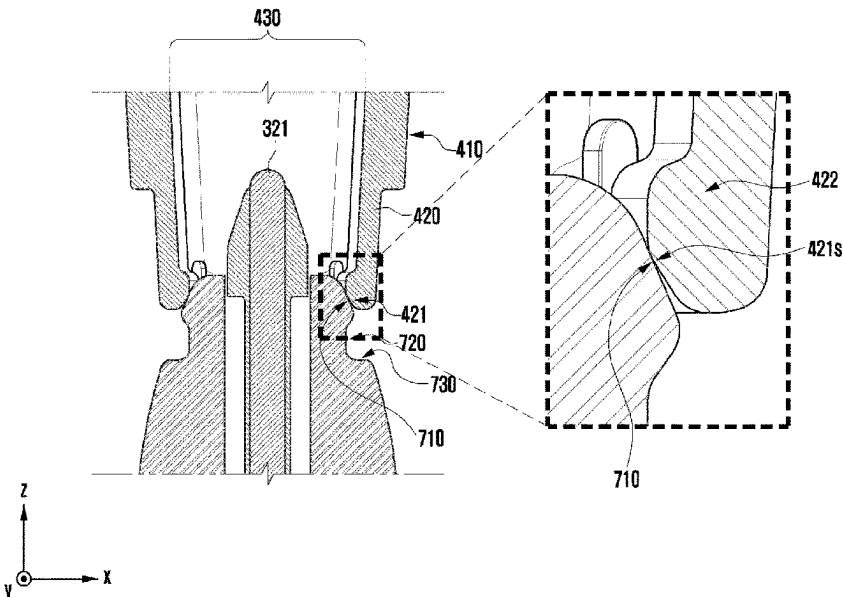
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(57) **ABSTRACT**

An electronic device may include: a housing; a hole formed in a portion of the housing; a storage space that is connected to the hole and extends into the housing; and a fastener that is disposed in the storage space and fastens an external electronic device, wherein the fastener comprises: a support part for fixing the fastener to the storage space; a coupler which extends from the support part and in the center of which an opening for inserting the external electronic device is formed; and two or more cutout parts formed by cutting out a portion of the coupler, wherein the cutout parts may form a repulsive force against an external force in an inward direction or an outward direction of the opening.

15 Claims, 16 Drawing Sheets



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FIG. 1

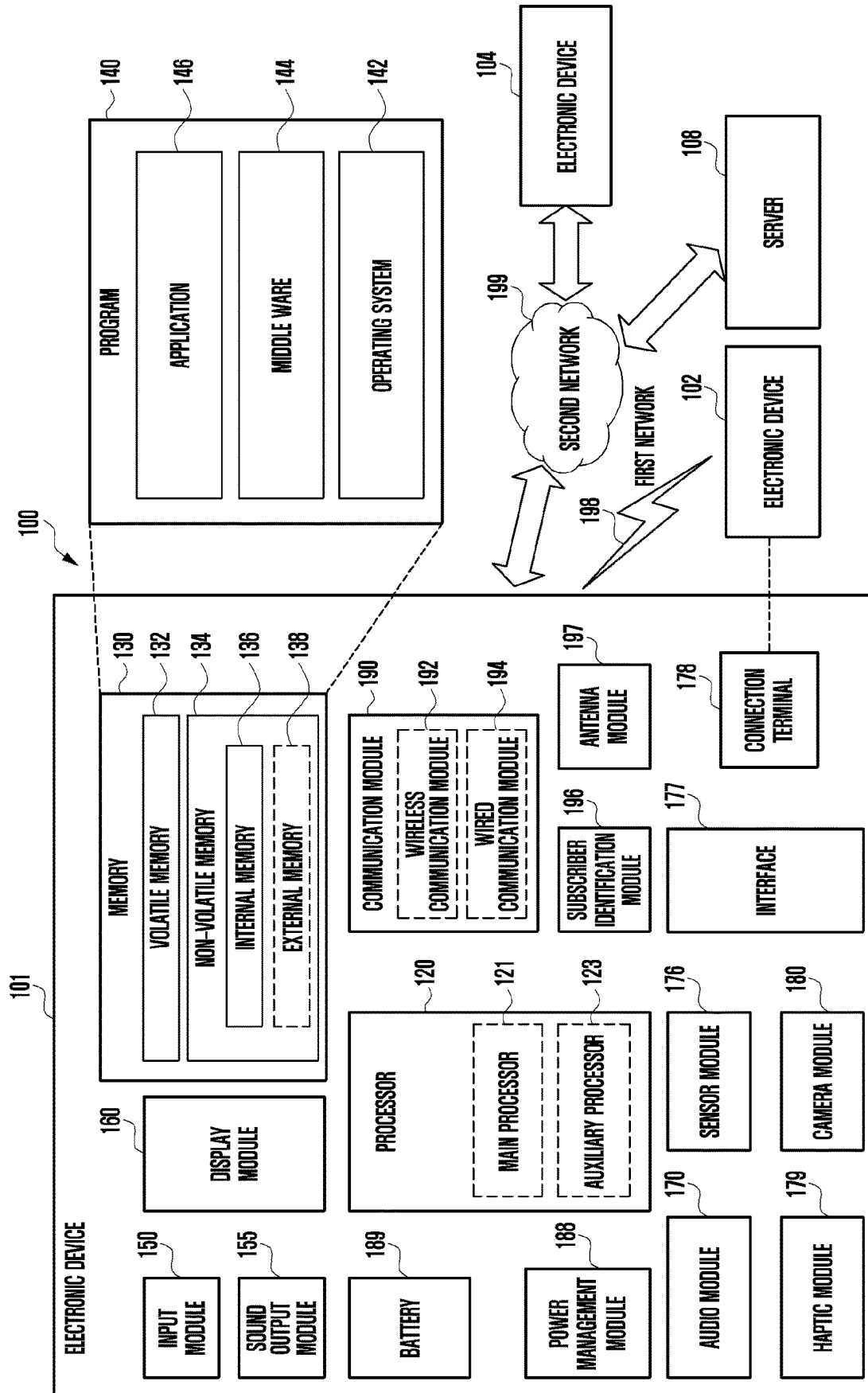


FIG. 2

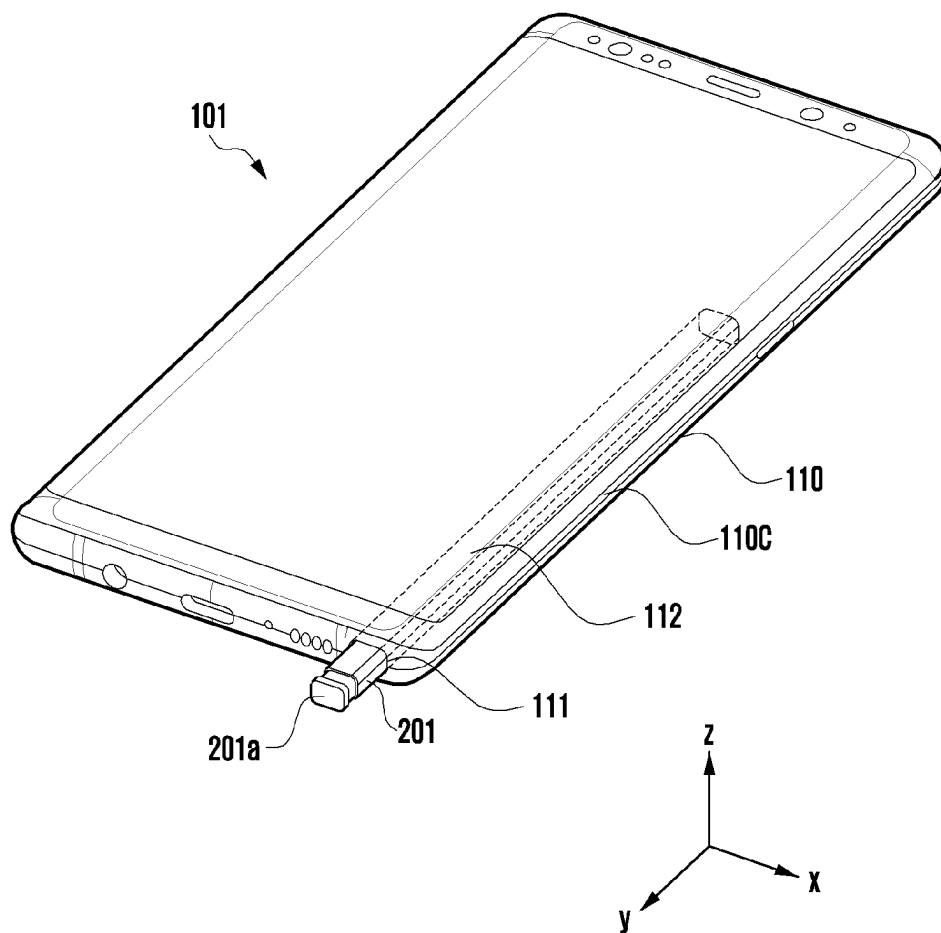


FIG. 3

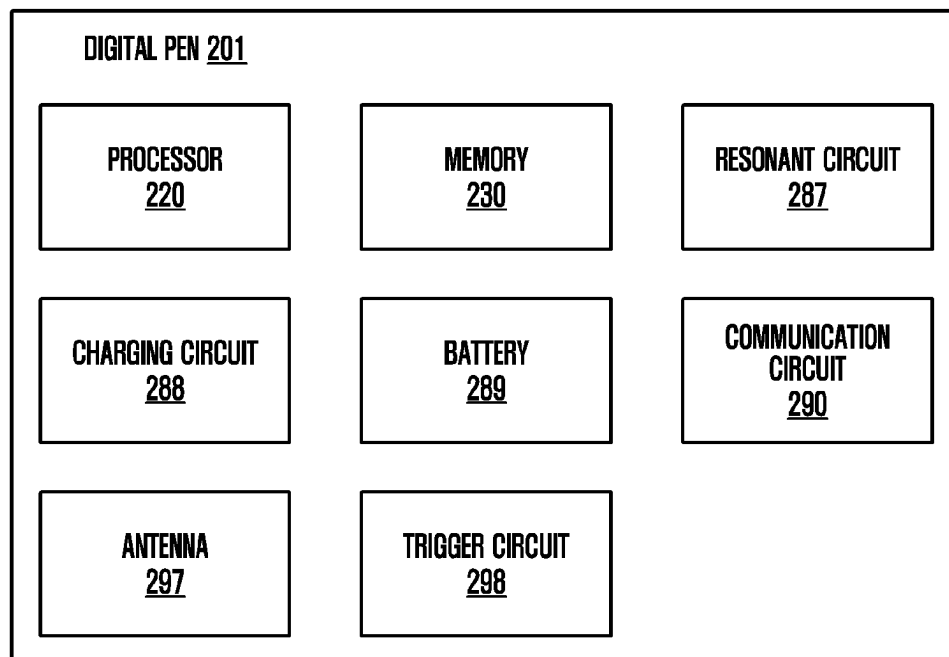


FIG. 4

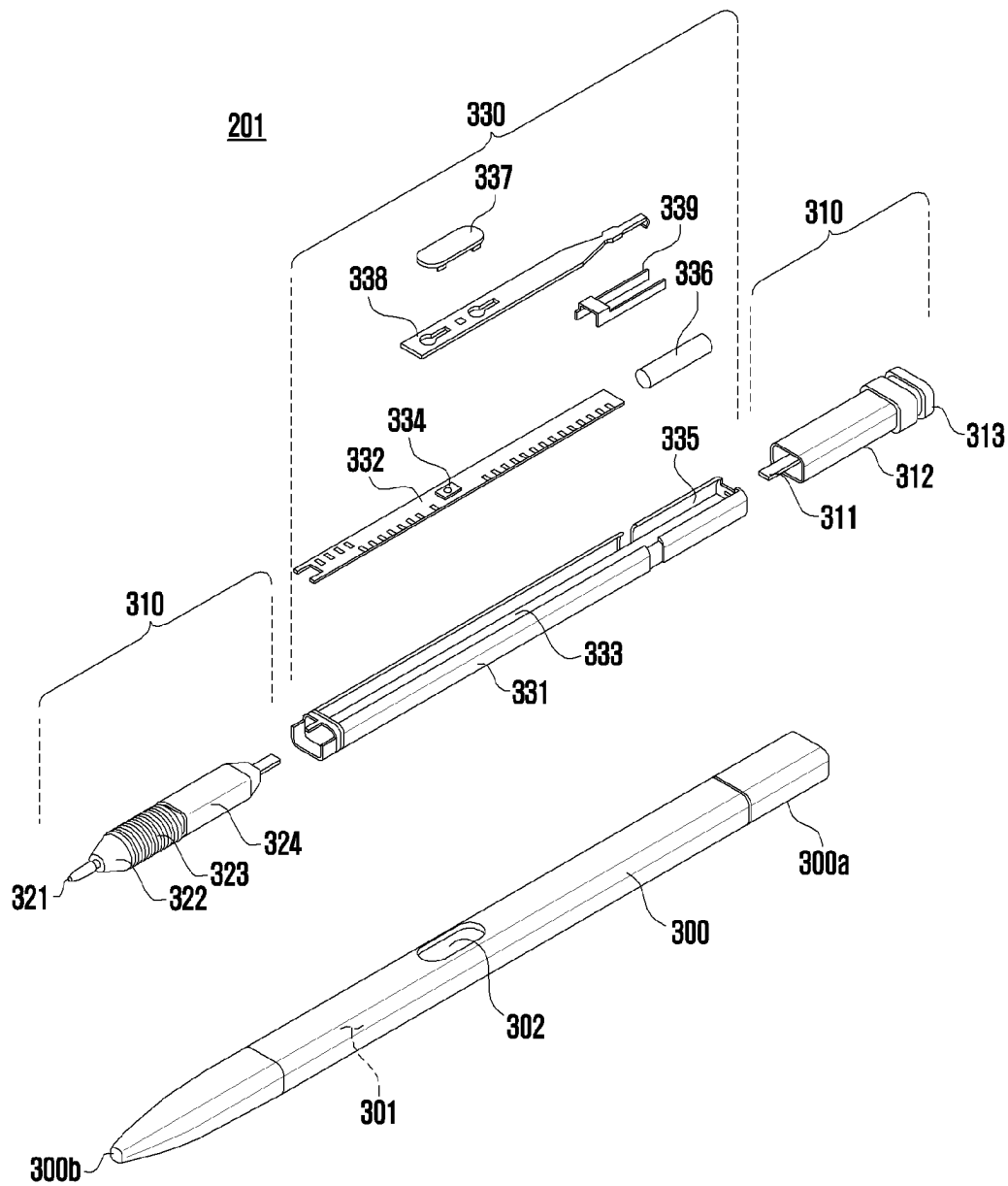


FIG. 5A

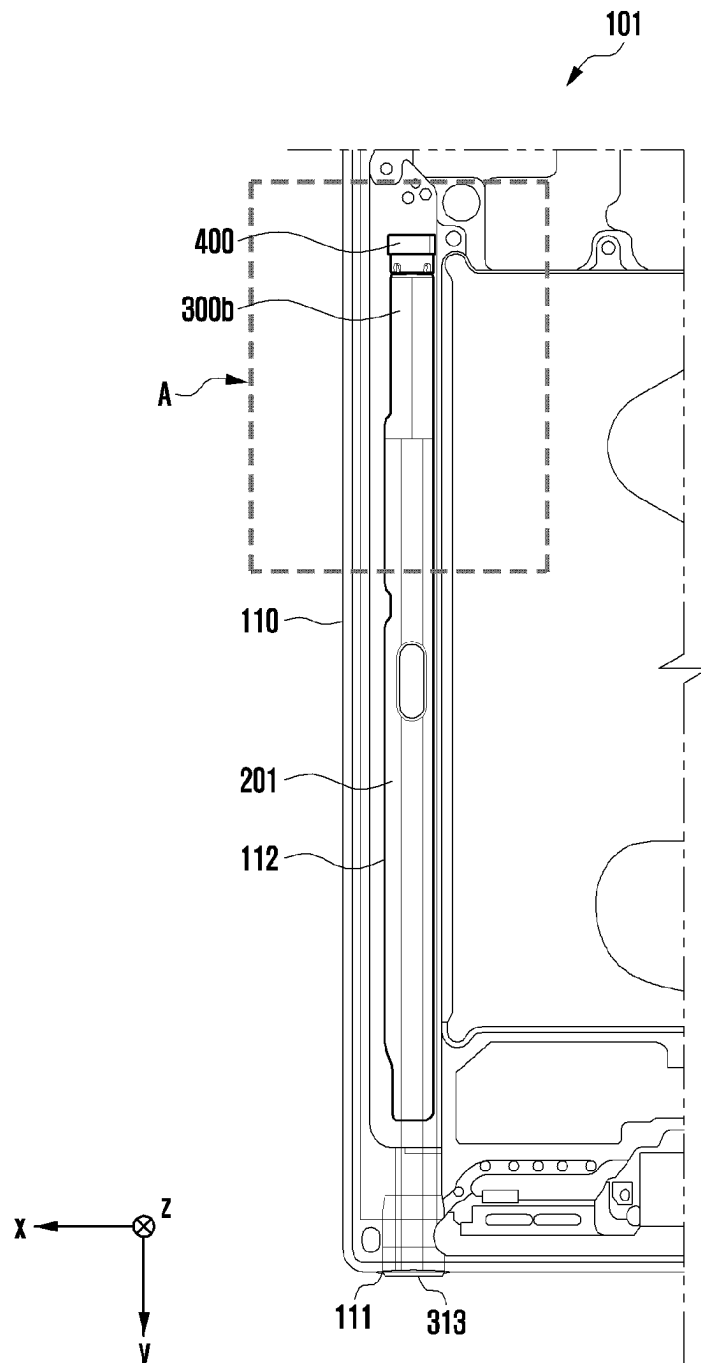


FIG. 5B

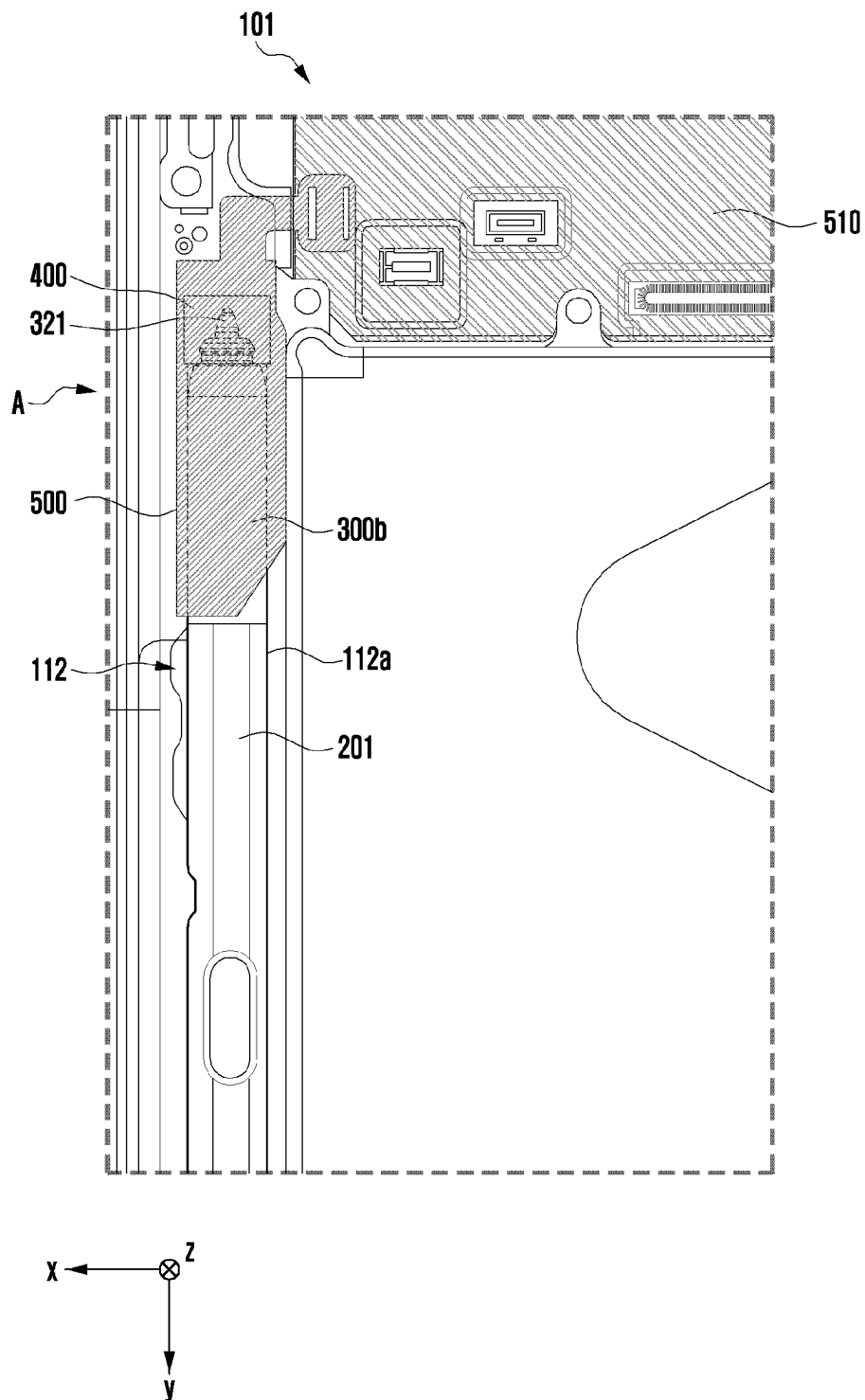


FIG. 5C

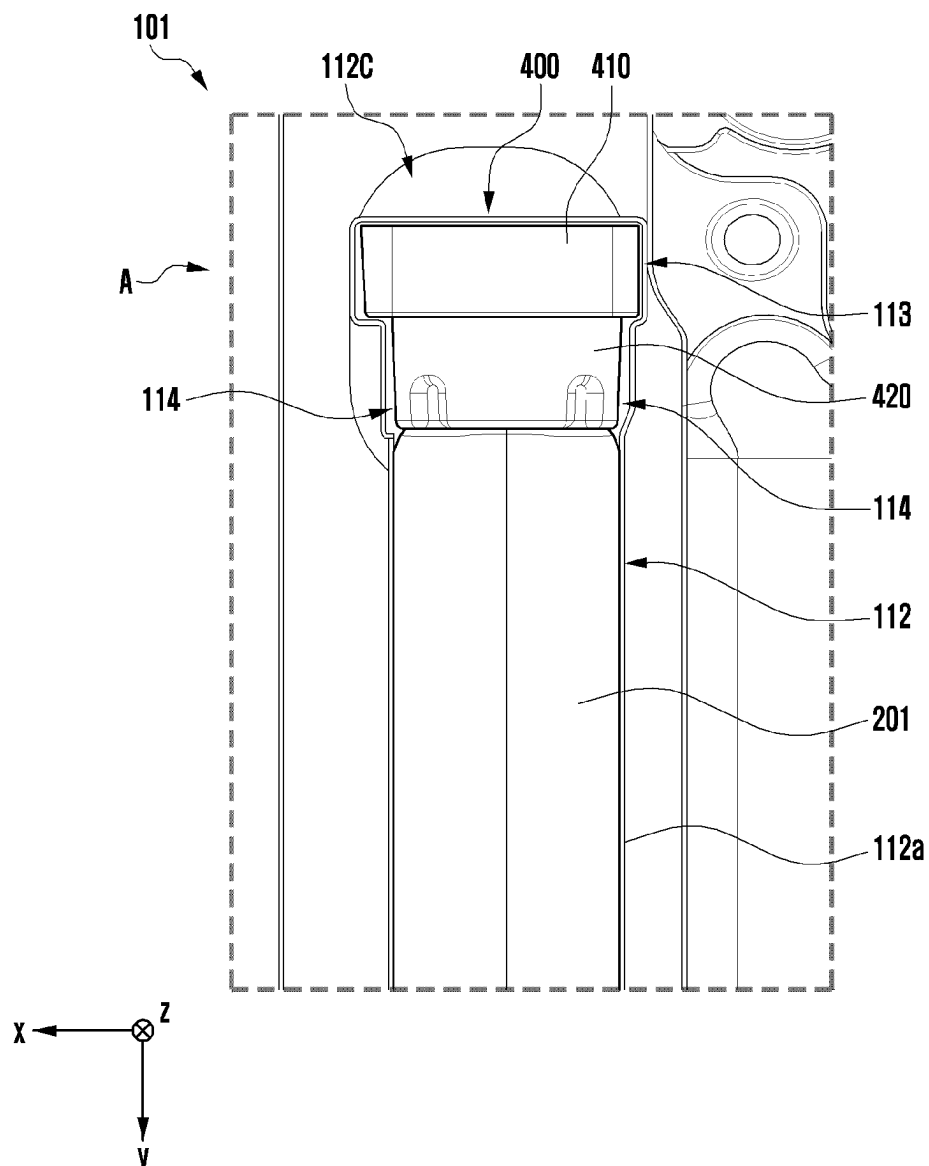


FIG. 6A

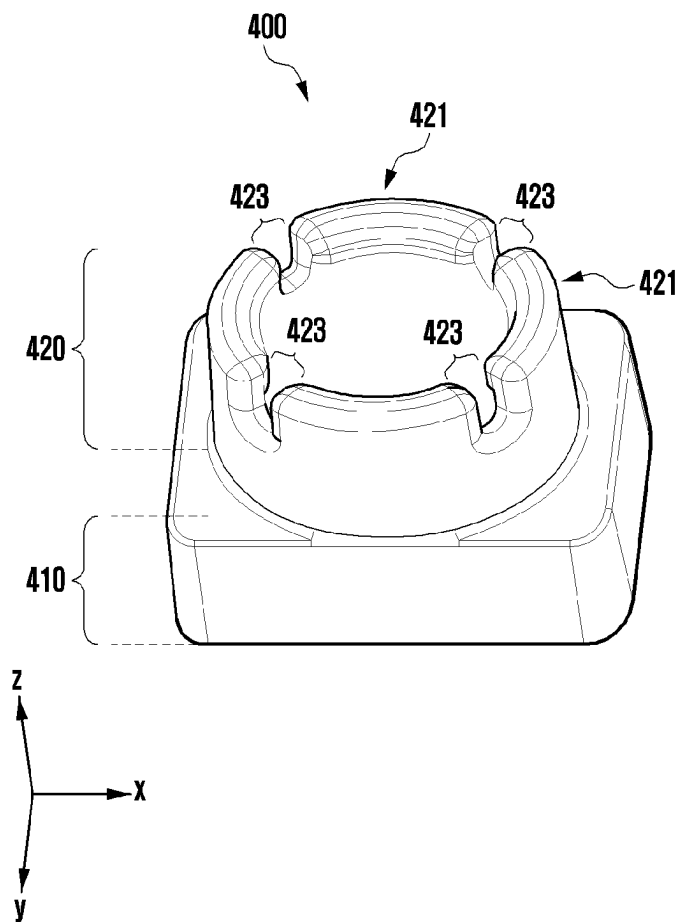


FIG. 6B

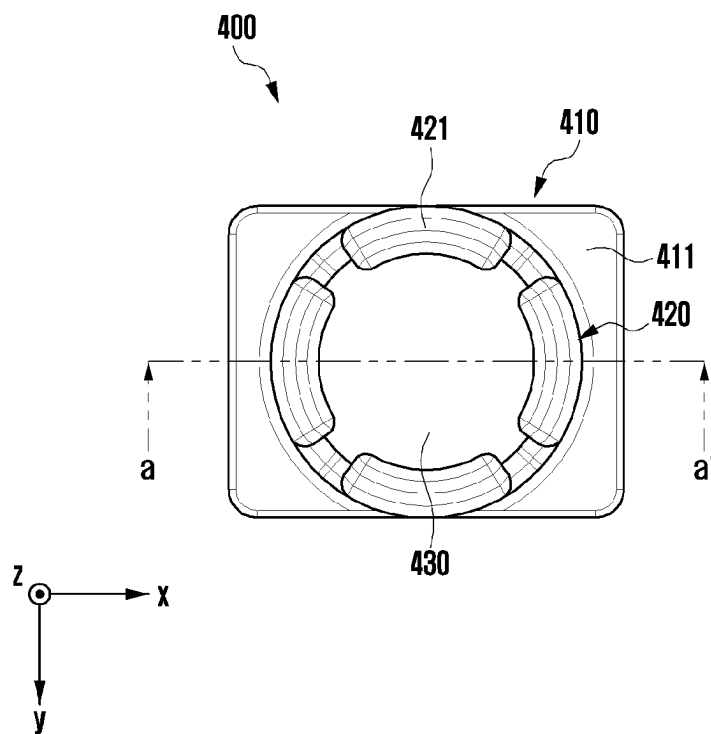


FIG. 6C

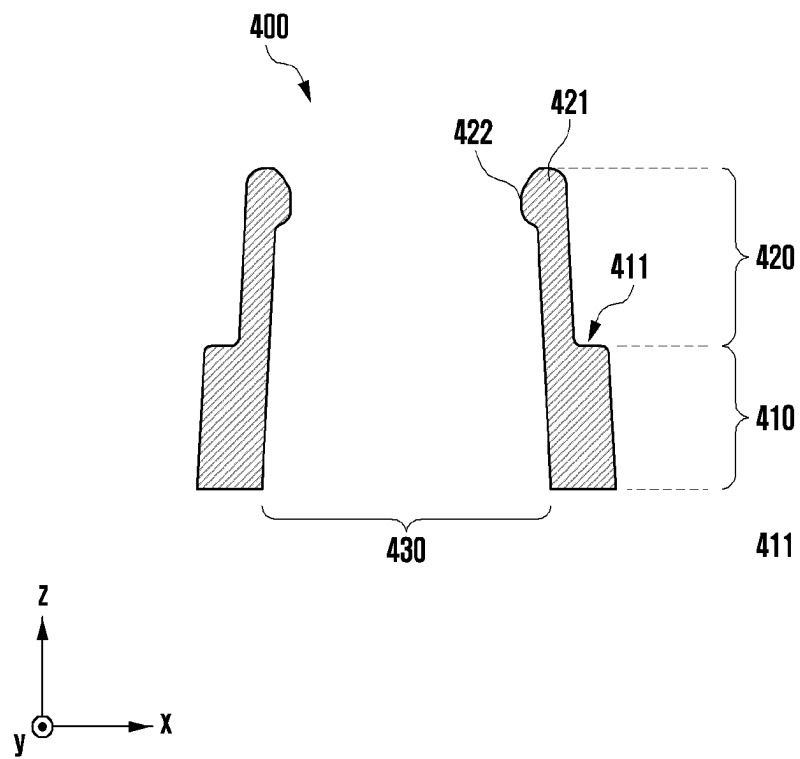


FIG. 6D

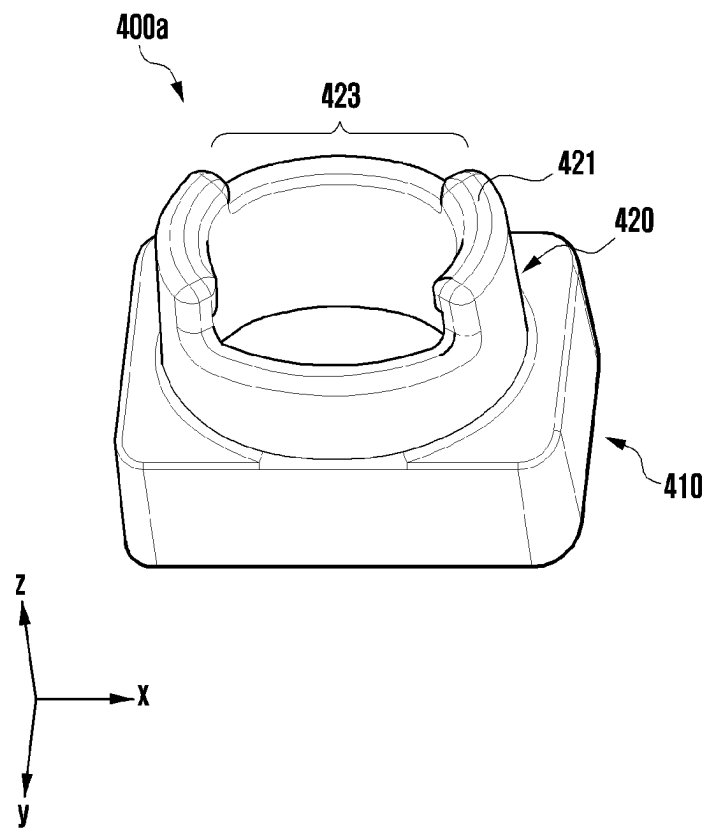


FIG. 7A

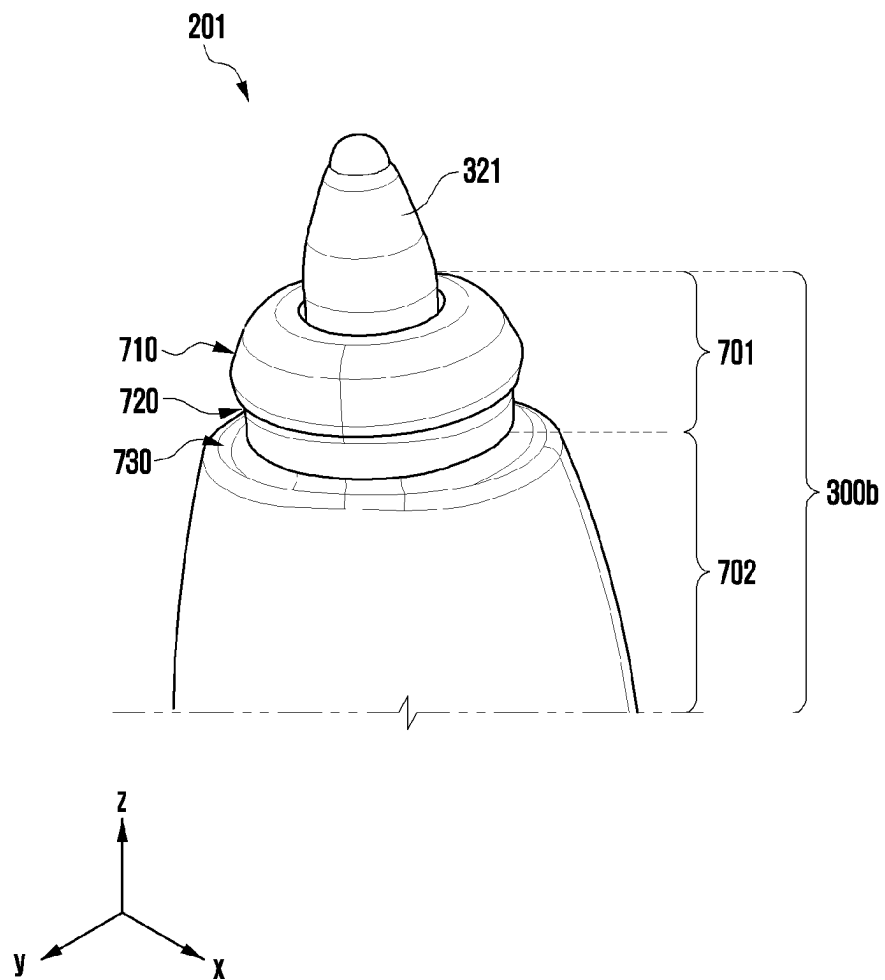


FIG. 7B

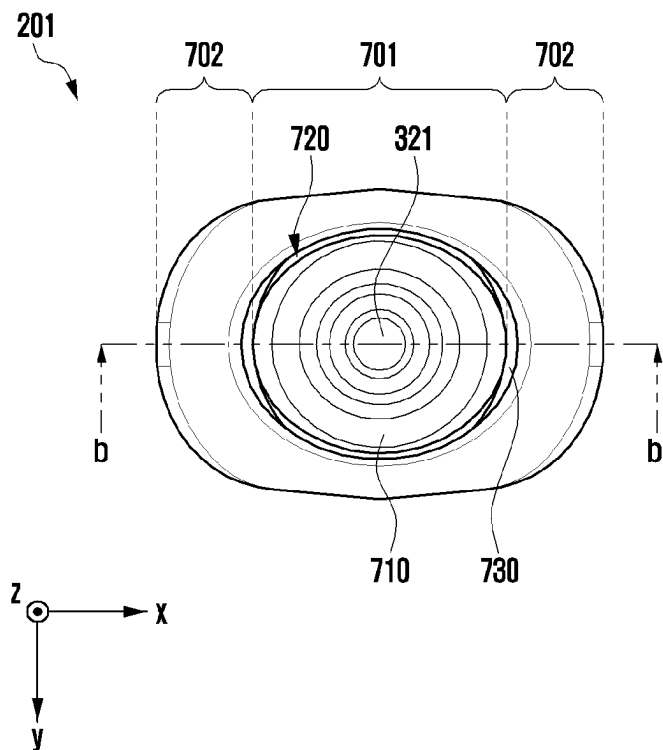


FIG. 7C

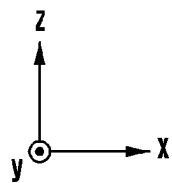
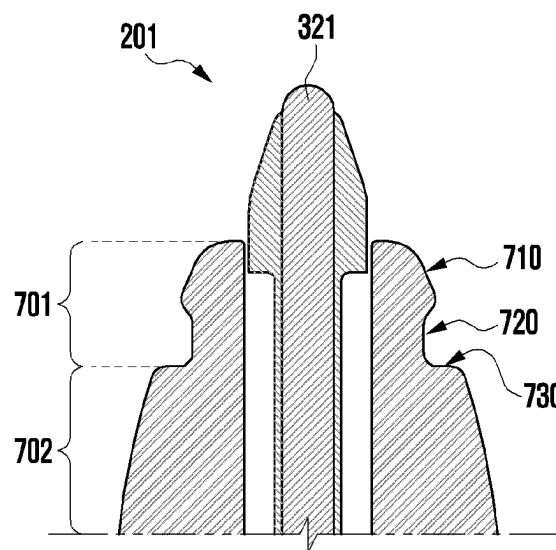
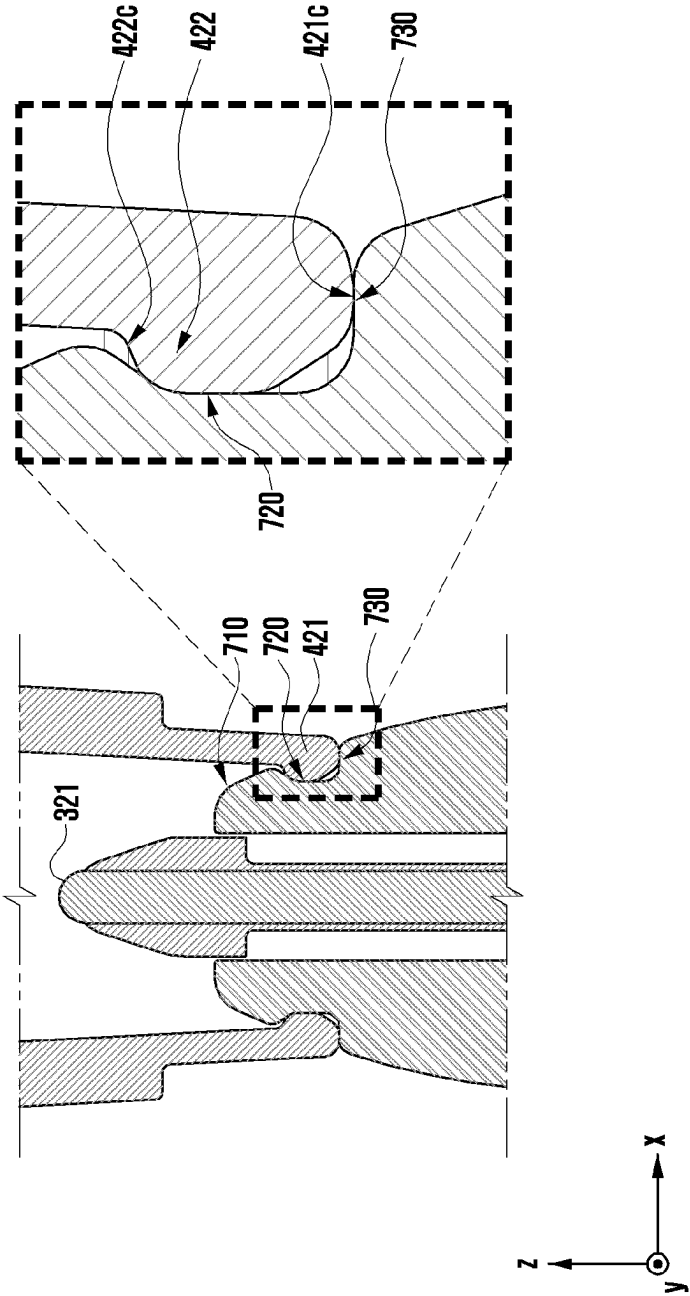


FIG. 8B



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ELECTRONIC DEVICE COMPRISING ELECTRONIC PEN STORAGE STRUCTURE, AND ELECTRONIC PEN

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of International Application No. PCT/KR2022/002452 filed on Feb. 18, 2022, designating the United States, in the Korean Intellectual Property Receiving Office, and claiming priority to KR Patent Application No. 10-2021-0023836 filed on Feb. 23, 2021, in the Korean Intellectual Property Office, the disclosures of all of which are hereby incorporated by reference herein in their entireties.

BACKGROUND

Field

Various example embodiments relate to an electronic device including a structure that is capable of storing an electronic pen and/or an electronic pen that is capable of being stored in the electronic device.

Description of Related Art

An electronic device may receive various inputs from the user by means of an input device (e.g., a digital pen) connected with the electronic device using wireless communication. The electronic device may receive input from a pen-type input device (hereinafter referred to as a “digital pen” for convenience).

Recently, various external electronic devices (e.g., a digital pen) have been developed that may be used as an adjunct to electronic devices. Accordingly, a structure capable of storing and/or fixing an external electronic device (e.g., a digital pen) outside or inside the electronic device may be formed.

SUMMARY

In a storage structure in which an electronic device stores a digital pen, a mounting space inside the electronic device may be utilized in case that the storage structure is formed outside the electronic device. In contrast, there is an effect of increasing portability of the digital pen in case that the storage structure is formed inside the electronic device. In case that the storage structure is formed inside the electronic device, the type of constituent elements that form the storage structure and a mounting position may affect utilization of a limited space inside a housing of the electronic device.

In addition, in case that the digital pen is stored inside the electronic device, but not aligned to be stored according to a structure in which the digital pen is inserted and fixed to the electronic device, e.g., an offcenter occurs, scratches on the electronic device or the digital pen may occur.

An electronic device according to various example embodiments may include: a housing; a hole formed in a portion of the housing; a storage space that is connected, directly or indirectly, to the hole and extends into the housing; and a fastening member that is disposed in the storage space and fastens an external electronic device, in which the fastening member may include: a support part for fixing the fastening member to the storage space; a coupling part which extends from the support part and in the center of which an opening for inserting the external electronic device

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is formed; and two or more cutout parts formed by cutting out a portion of the coupling part, in which the cutout parts may form a repulsive force against an external force in an inward direction or an outward direction of the opening.

5 An external electronic device according to various example embodiments may include: a housing; a first end formed at one end of the housing; and a second end formed at the other end of the housing in a direction opposite to the first end, and in which the second end may include: an inclined portion forming a slope on a surface of the pen housing by increasing in diameter in a direction from the second end toward the first end; a catching groove formed at an end of the inclined portion in a direction toward the first end, and including a diameter smaller than a maximum diameter formed by the inclined portion; and a planar portion extending from the catching groove and configured to interrupt the catching groove in a direction perpendicular to the diameter.

According to various example embodiments, the digital pen may be aligned and stored in the electronic device to prevent or reduce damage to the electronic device and/or the digital pen.

According to various example embodiments, it is possible to increase utilization of mounting space inside the electronic device.

BRIEF DESCRIPTION OF DRAWINGS

In connection with the description of the drawings, the same or similar reference numerals may be used for the same or similar components. The above and other aspects, features and advantages of certain embodiments of the present disclosure will be more apparent from the following detailed description, taken in conjunction with the accompanying drawings, in which:

FIG. 1 is a block diagram of an electronic device in a network environment according to various example embodiments.

FIG. 2 is a perspective view illustrating the electronic device according to an example embodiment, including a digital pen.

FIG. 3 is a block diagram illustrating the digital pen according to an example embodiment.

FIG. 4 is an exploded perspective view illustrating the digital pen according to an example embodiment.

FIG. 5A is a view illustrating a structure for mounting a digital pen according to various example embodiments.

FIGS. 5B and 5C are enlarged views illustrating a partial area of FIG. 5A.

FIG. 6A is a perspective view illustrating the fastening member according to various example embodiments.

FIG. 6B is a top view illustrating the fastening member according to various example embodiments.

FIG. 6C is a cross-sectional view illustrating the fastening member according to various example embodiments.

FIG. 6D is a perspective view illustrating the fastening member according to various example embodiments.

FIG. 7A is a perspective view illustrating a second end of the digital pen according to various example embodiments.

FIG. 7B is a top view illustrating the digital pen according to various example embodiments.

FIG. 7C is a cross-sectional view illustrating the second end of the digital pen according to various example embodiments.

FIG. 8A is a cross-sectional view illustrating a process of coupling the digital pen with the fastening member according to various example embodiments.

FIG. 8B is a cross-sectional view illustrating that the digital pen and fastening member are coupled according to various example embodiments.

DETAILED DESCRIPTION

FIG. 1 is a block diagram illustrating an electronic device **101** in a network environment **100** according to various embodiments. Referring to FIG. 1, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to an embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication

processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal

or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

The power management module **188** may manage power supplied to the electronic device **101**. According to an embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™ wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy

cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN)). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit

board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

FIG. 2 is a perspective view illustrating the electronic device according to an embodiment, including a digital pen.

FIG. 3 is a block diagram illustrating the digital pen according to an embodiment.

FIG. 4 is an exploded perspective view illustrating the digital pen according to an embodiment.

With reference to FIG. 2, the electronic device **101** according to an embodiment may include the constitution illustrated in FIG. 1, and may include a structure into which a digital pen **201** (e.g., a stylus pen) may be inserted. The electronic device **101** includes a housing **110**, and a portion of the housing, such as a portion of a side surface **110C**, may include a hole **111**. The electronic device **101** may include a storage space **112** connected, directly or indirectly, to the hole **111**, and the digital pen **201** may be inserted in the storage space **112**. According to the illustrated embodiment, the digital pen **201** may include, at one end thereof, a pressable button **201a** to facilitate removal of the digital pen **201** from the storage space **112** of the electronic device **101**. When the button **201a** is pressed, a repulsion mechanism (e.g., at least one spring) that is constituted in conjunction

with the button **201a** may be activated to separate the digital pen **201** from the storage space **112**.

With reference to FIG. 3, the digital pen **201** according to an embodiment may include a processor **220**, a memory **230**, a resonant circuit **287**, a charging circuit **288**, a battery **289**, a communication circuit **290**, an antenna **297**, and/or a trigger circuit **298**. In some embodiments, the processor **220**, at least a portion of the resonant circuit **287**, and/or at least a portion of the communication circuit **290** of the digital pen **201** may be constituted on a printed circuit board or in the form of a chip. The processor **220**, resonant circuit **287**, and/or communication circuit **290** may be electrically connected, directly or indirectly, with the memory **230**, charging circuit **288**, battery **289**, antenna **297**, or trigger circuit **298**. The digital pen **201** according to an embodiment may be constituted by only the resonant circuit and a button.

The processor **220** may include a generic processor constituted to execute a customized hardware module or software (e.g., an application program). The processor may include hardware constituent elements (functions) or software elements (programs) that include at least one of various sensors provided in the digital pen **201**, a data measurement module, an input/output interface, a module to manage a state or environment of the digital pen **201**, or a communication module. The processor **220** may include one or more combinations of hardware, software, or firmware, for example. According to an embodiment, the processor **220** may receive a proximity signal corresponding to an electromagnetic field signal generated by a digitizer **160** of the electronic device **101** over a resonant circuit **287**. Once the proximity signal is confirmed, the resonant circuit **287** may be controlled to transmit an electro-magnetic resonance (EMR) input signal to the electronic device **101**.

The memory **230** may store information related to an operation of the digital pen **201**. For example, the information may include information on communicating with the electronic device **101** and frequency information related to an input operation of the digital pen **201**.

The resonant circuit **287** may include at least one of a coil, an inductor, or a capacitor. The resonant circuit **287** may be utilized that the digital pen **201** generates a signal including a resonant frequency. For example, to generate the signal, the digital pen **201** may utilize at least one of an electro-magnetic resonance (EMR) method, an active electrostatic (AES) method, or an electrically coupled resonance (ECR) method. In case that the digital pen **201** transmits a signal by EMR method, the digital pen **201** may generate a signal including a resonant frequency based on an electromagnetic field generated by an inductive panel of the electronic device **101**. In case that the digital pen **201** transmits a signal by the AES method, the digital pen **201** may utilize capacity coupling with the electronic device **101** to generate the signal. In case that the digital pen **201** transmits a signal by the ECR method, the digital pen **201** may generate a signal including a resonant frequency based on an electric field generated by a capacitive device of the electronic device. According to an embodiment, the resonant circuit **287** may be utilized to change intensity or frequency of an electromagnetic field based on a user's operational state. For example, the resonant circuit **287** may provide a frequency to recognize a hovering input, a drawing input, a button input, or an erasing input.

In case that the charging circuit **288** is connected, directly or indirectly, to the resonant circuit **287** based on a switching circuit, a resonant signal generated by the resonant circuit **287** may be rectified into a direct current signal and provided to the battery **289**. According to an embodiment, the digital

pen **201** may utilize a voltage level of a direct current signal detected by the charging circuit **288** to determine whether the digital pen **201** has been inserted into the electronic device **101**.

The battery **289** may be constituted to store power required to operate the digital pen **201**. The battery may include, for example, a lithium-ion battery, or a capacitor, and may be rechargeable or replaceable. According to an embodiment, the battery **289** may be charged using power (e.g., a direct current signal (DC power)) provided by the charging circuit **288**.

The communication circuit **290** may be constituted to perform a wireless communication function between the digital pen **201** and the communication module **190** of the electronic device **101**. According to an embodiment, the communication circuit **290** may transmit state information and input information of the digital pen **201** to the electronic device **101** using a short-range communication method. For example, the communication circuit **290** may transmit to the electronic device **101** direction information (e.g., motion sensor data) on the digital pen **201** obtained by the trigger circuit **298**, voice information inputted by a microphone, or battery level information on the battery **289**. In an example, the short-range communication method may include at least one of Bluetooth, Bluetooth low energy (BLE), or wireless LAN.

The antenna **297** may be utilized to transmit signals or power to, or receive from, an external source (e.g., the electronic device **101**). According to an embodiment, the digital pen **201** may include a plurality of antennas **297**, of which at least one antenna **297** may be selected that is suitable for a communication method. The communication circuit **290** may exchange signals or power with an external electronic device via the selected at least one antenna **297**.

The trigger circuit **298** may include at least one button or sensor circuit. According to an embodiment, the processor **220** may identify an input method (e.g., touch or press) or type (e.g., EMR button or BLE button) of a button on the digital pen **201**. According to an embodiment, the sensor circuit may generate electrical signals or data values that correspond to an internal operational state of the digital pen **201** or an external environmental state. For example, the sensor circuit may include at least one of a motion sensor, a battery level detection sensor, a pressure sensor, a light sensor, a temperature sensor, a geomagnetic sensor, and a biometric sensor. According to an embodiment, the trigger circuit **298** may utilize an input signal from a button or a signal from a sensor to transmit a trigger signal to the electronic device **101**.

With reference to FIG. 4, the digital pen **201** may include a pen housing **300**, which constitutes an outline of the digital pen **201**, and an inner assembly inside the pen housing **300**. In the illustrated embodiment, the inner assembly, including all of the various components that are mounted inside the pen, may be inserted inside the pen housing **300** in a single assembly operation.

The pen housing **300** may have an elongated shape between a first end **300a** and a second end **300b**, and may include the storage space **112** therein. The pen housing **300** may be elliptical in cross-section with a long axis and a short axis, and may be formed overall in the shape of an elliptical column. The storage space **301** of the electronic device **101** may also be formed elliptically in cross-section, corresponding to the shape of the pen housing **300**. The pen housing **300** may include a synthetic resin (e.g., plastic) and/or a metallic material (e.g., aluminum). According to an embodi-

ment, the second end **300b** of the pen housing **300** may be constituted of a synthetic resin material.

The inner assembly may have an elongated shape corresponding to the shape of the pen housing **300**. The inner assembly may be divided into three constituents along the lengthwise direction. For example, the inner assembly may include an ejection member **310** disposed in a position corresponding to the first end **300a** of the pen housing **300**, a coil part **320** disposed in a position corresponding to the second end **300b** of the pen housing **300**, and a circuit board part **330** disposed in a position corresponding to a body of the housing.

The ejection member **310** may include a constituent configured to eject the digital pen **201** from the storage space **112** of the electronic device **101**. According to an embodiment, the ejection member **310** may include a shaft **311**, an ejection body **312** disposed around the perimeter of the shaft **311** and forming an overall outline of the ejection member **310**, and a button part **313**. When the inner assembly is fully inserted into the pen housing **300**, a portion including the shaft **311** and the ejection body **312** may be enclosed by the first end **300a** of the pen housing **300**, and the button part **313** (e.g., **201a** in FIG. 2) may be exposed to the outside of the first end **300a**. A plurality of components not illustrated, such as cam members or resilient members, may be disposed within the ejection body **312** to form a push-pull structure. In an embodiment, the button part **313** may be coupled substantially with the shaft **311** to perform a linear reciprocating motion relative to the ejection body **312**. According to various embodiments, the button part **313** may include a button with a catch structure formed to allow a user to pull out the digital pen **201** using a fingernail. According to an embodiment, the digital pen **201** may include a sensor to detect the linear reciprocating motion of the shaft **311**, thereby providing another input method.

The coil part **320** may include a pen tip **321** exposed to the outside of the second end **300b** when the inner assembly is fully inserted into the pen housing **300**, a packing ring **322**, a plurally wound coil **323**, and/or a pen pressure sensing part **324** configured to obtain a change in pressure depending on pressurization of the pen tip **321**. The packing ring **322** may include epoxy, rubber, urethane, or silicone. The packing ring **322** may be provided for waterproofing and dustproofing purposes, and may protect the coil part **320** and the circuit board part **330** from water immersion or dust. According to an embodiment, the coil **323** may form a resonant frequency in a configured frequency band (e.g., 500 kHz), and may be combined with at least one element (e.g., a capacitive element (capacitor)) to modulate the resonant frequency formed by the coil **323** in some range.

The circuit board part **330** may include a printed circuit board **332**, a base **331** that surrounds at least one surface of the printed circuit board **332**, and an antenna. According to an embodiment, a board resting part **333** is formed on an upper surface of the base **331** on which a printed circuit board **332** is disposed, and the printed circuit board **332** may be fixed in a resting state on the board resting part **333**. According to an embodiment, the printed circuit board **332** may include a top surface and a bottom surface, and a variable capacitance capacitor or switch **334** connecting to the coil **323** may be disposed on the top surface, and the charging circuit, battery, or communication circuit may be disposed on the bottom surface. The battery may include an electric double layered capacitor (EDLC). The charging circuit is positioned between the coil **323** and the battery, and may include voltage detector circuitry and a rectifier.

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The antenna may include an antenna structure **339** such as the example illustrated in FIG. 4 and/or an antenna embedded in the printed circuit board **332**. According to various embodiments, the switch **334** may be provided on the printed circuit board **332**. A side button **337** provided on the digital pen **201** is utilized to push the switch **334** and may be exposed to the outside through a side surface opening **302** of the pen housing **300**. The side button **337** is supported by a support member **338** and may be restored or maintained in a state in which the side button **337** is disposed in a predetermined position by the support member **338** providing elastic resilience when there is no external force acting on the side button **337**.

The circuit board part **330** may include other packing rings, such as an O-ring. For example, an O-ring made of an elastic material may be disposed at each end of the base **331** to form a sealing structure between the base **331** and the pen housing **300**. In some embodiments, the support member **338** may tightly fit against an inner wall of the pen housing **300**, partially around the side surface opening **302**, to form a sealing structure. For example, the circuit board part **330** may also form a waterproof, dustproof structure similar to the packing ring **322** of the coil part **320**.

The digital pen **201** may include a battery resting part **335** where a battery **336** is disposed on an upper surface of the base **331**. The battery **336** that may be mounted in the battery resting part **335** may include, for example, a cylinder type battery.

The digital pen **201** may include a microphone (not illustrated). The microphone may be connected directly to the printed circuit board **332**, or may be connected, directly or indirectly, to a separate flexible printed circuit board (FPCB) (not illustrated) that is connected to the printed circuit board **332**. According to various embodiments, the microphone may be disposed in a position parallel to the side button **337** in the longitudinal direction of the digital pen **201**.

FIG. 5A is a view illustrating a structure for mounting a digital pen according to various embodiments.

FIGS. 5B and 5C are enlarged views illustrating a partial area of FIG. 5A.

With reference to FIGS. 5A, 5B, and 5C, a mounting structure including at least the hole **111**, the storage space **112**, and a fastening member/fastener **400** may be formed to mount the digital pen **201** inside the housing **110** of the electronic device **101**.

According to various embodiments, the housing **110** may form an opening, such as the hole **111**, on one side surface thereof. According to an embodiment, the hole **111** may form a size substantially equal to the cross-section of the digital pen **201**. According to an embodiment, a diameter of the hole **111** may be equal to or greater than a maximum or large diameter of the digital pen **201**.

According to various embodiments, the storage space **112** may be a space formed to extend from the hole **111** to a direction towards the inside of the housing **110** (e.g., in the opposite direction of the y-axis). According to an embodiment, most of an area of the storage space **112** (e.g., a connecting passage **112a**) may be a space that an opening having the same diameter as the hole **111** extends through and forms. For example, a diameter of the connecting passage **112a** may be the same as a diameter of the hole **111**. According to an embodiment, a size of the diameter formed by the connecting passage **112a** may be equal to or greater than a maximum or large diameter formed by the digital pen **201**. According to an embodiment, a length that the storage

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space **112** has (e.g., a length of the y-axis direction) may be equal to or greater than a length formed by the digital pen **201**.

According to various embodiments, the digital pen **201** may be inserted inside the electronic device **101**. According to various embodiments, the digital pen **201** may be inserted into the electronic device **101** in a specific direction. According to various embodiments, at least one of the both ends of the digital pen **201** (e.g., the first end **300a** or the second end **300b** in FIG. 4) may be inserted into the electronic device **101**. According to an embodiment, the digital pen **201** may be inserted into the electronic device **101** beginning from the second end **300b**.

According to various embodiments, the digital pen **201** may be inserted into the electronic device **101** from the hole **111** and stored in the storage space **112**. According to an embodiment, the digital pen **201** may be slidably inserted into the storage space **112** according to a structure formed by the storage space **112**. According to various embodiments, a portion of the digital pen **201** that has been inserted (e.g., a portion of the button part **313**) may protrude outside of the hole **111**. According to an embodiment, the protruding button part **313** may reciprocate relative to the ejection body (e.g., the ejection member **312** in FIG. 4) when pressurized, and a catching structure included in the button part **313** may be exposed outside of the housing **110** to catch and eject the digital pen **201**.

According to various embodiments, the electronic device **101** may identify whether the digital pen **201** has been inserted into the electronic device **101**. According to an embodiment, the electronic device **101** may radiate an electromagnetic field and/or an electric field to the digital pen **201**. With reference to FIG. 5B, the electronic device **101** may include at least one printed circuit board (e.g., a first printed circuit board **510** and a second printed circuit board **500**). According to an embodiment, the first printed circuit board **510** is electrically connected, directly or indirectly, to the second printed circuit board **500** and a battery (e.g., the battery **189** in FIG. 1), and the second printed circuit board **500** may receive power from the battery **189** via the first printed circuit board **500**. According to an embodiment, the second printed circuit board **500** may output an electromagnetic field signal and/or an electric field signal to resonate with the digital pen **201**. According to an embodiment, the second printed circuit board may be disposed at a position corresponding to at least a portion of the digital pen **201** (e.g., the second end **300b**) in case that the digital pen **201** is stored. According to an embodiment, the second printed circuit board **500** may generate a resonant signal by the digital pen **201** in case that the digital pen **201** is inserted, and the electronic device **101** may detect the resonant signal to identify whether the digital pen **201** is inserted.

According to various embodiments, the digital pen **201** may be inserted and fixed inside the storage space **112**. With reference to FIG. 5C, the fastening member/fastener **400** may be disposed in at least a portion of the storage space **112**. According to various embodiments, the digital pen **201** may be fixed inside the storage space **112** by being at least partially engaged with the fastening member **400**. According to an embodiment, the fastening member **400** may fix the digital pen **201** to the storage space **112**. According to various embodiments, the fastening member **400** may be disposed inside the housing **110** such that a center thereof is aligned with the digital pen **201**. According to an embodiment, the storage space **112** may include the communicating part **112a**, which is formed to extend so that the digital pen

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112 may be inserted, and a mounting area 112c, in which the fastening member 400 may be mounted. According to an embodiment, the fastening member 400 may be mounted in the mounting area 112c. According to an embodiment, the mounting area 112c is formed in the form of a groove and may include interruption part 113 that interrupts at least some area of the fastening member 400. According to various embodiments, the fastening member 400 may include a support part 410 that fixes the fastening member 400 inside the mounting area 112c and a coupling part/coupler 420 that is coupled to the digital pen 201. According to an embodiment, the support part 410 may have a larger diameter than the coupling part/coupler 420, and may form a stepped structure (e.g., a stepped part 411 in FIG. 6C) that is formed by a diameter difference from the coupling part 420. According to an embodiment, the stepped part formed by the support part 410 (e.g., the stepped part 411 in FIG. 6C) may be fixed inside the mounting area 112c with being interrupted by the interruption part 113 included in the mounting area 112c. According to an embodiment, the mounting area 112c may include a clearance 114 to include a range in which a portion of the fastening member 400 (e.g., the coupling part 420) deforms. According to an embodiment, the coupling part 420 may spread in a left-right direction (e.g., along the x-axis and/or opposite the x-axis) while coupling with the digital pen 201, and may extend in a direction in which the clearance 114 is formed. According to an embodiment, the fastening member 400 may be disposed to be aligned in one direction (e.g., in the y-axis direction) when the digital pen 201 is inserted. Therefore, the digital pen 201 may be aligned without an offcenter and inserted inside the electronic device 101.

FIG. 6A is a perspective view illustrating the fastening member 400 according to various embodiments.

FIG. 6B is a top view illustrating the fastening member 400 according to various embodiments.

FIG. 6C is a cross-sectional view illustrating the fastening member 400 according to various embodiments.

FIG. 6D is a perspective view illustrating the fastening member 400 according to various embodiments.

FIG. 6C may be a cross-sectional view in which the fastening member 400 is cut along line aa' (e.g., the line aa' in FIG. 6B).

With reference to FIGS. 6A, 6B, and 6C, the fastening member 400 may include the support part 410 and the coupling part 420.

The fastening member/fastener 400 may be fixed to the storage space 112 by the support part 410. According to various embodiments, the fastening member 400 may be mounted and fixed to some area (e.g., the mounting area 112c in FIG. 5C) of the storage space (e.g., the storage space 112 in FIG. 5C). According to an embodiment, a diameter of the support part 410 may be less than or equal to the mounting area 112c in the storage space 112, and may be greater than other areas of the storage space 112 (e.g., the connecting passage 112a in FIG. 5C). According to an embodiment, the stepped part 411 formed on the support part 410 may be fixed inside the mounting area 112c with being interrupted by the interruption part (e.g., the interruption part 113 in FIG. 5C) formed by a difference in diameters of the mounting area 112c and the connecting passage 112a.

The coupling part/coupler 420 may fix the digital pen (e.g., the digital pen 201 in FIG. 2). According to an embodiment, the coupling part 420 may include a structure protrudingly extending from the support part 410 and including a circular and/or substantially circular opening 430 therein into which the digital pen 201 is inserted.

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According to an embodiment, the opening 430 may include a structure that is circular and/or substantially circular in cross-section, and may be formed continuously from the coupling part 420 to the support part 410. According to an embodiment, the digital pen 201 may be inserted into the fastening member/fastener 400 through the opening 430. According to an embodiment, the cross-section of the opening 430 may be a shape corresponding to cross-section of a top portion of the digital pen 201 (e.g., a top portion 701 in FIG. 7A). According to an embodiment, a maximum or large diameter of the cross-section of the opening 430 may be greater than or equal to a maximum or large diameter formed by an area of the top portion 701 of the digital pen 201. According to various embodiments, the digital pen 210 may be inserted from the opening 430 and coupled to the coupling part 420. According to various embodiments, the coupling part 420 may include a cutout structure including at least two cutout parts 421. According to an embodiment, the cutout part 421 may be a portion formed by a cutout clearance 423 that is cut out in some area of the coupling part 420. According to various embodiments, the cutout part 421 may form a repulsive force against an external force. According to an embodiment, the cutout part 421 may form a repulsive force in an outward direction (e.g., in a direction opposite the opening 430) against an external force acting inwardly, that is, in a direction toward the opening 430. According to an embodiment, the cutout part 421 may apply a repulsive force inwardly against an external force acting outwardly, that is, in the opposite direction of the opening 430. According to an embodiment, the cutout part 421 may be bent and deformed by an external force in a direction of the external force. Therefore, elastic and/or tensile forces may be applied by deformation to apply a repulsive force in the opposite direction of the external force. According to an embodiment, the cutout part 421 may include a catching projection 422 protruding from a portion thereof. The catching projection 422 may be formed at some area of the cutout part 421, such as at an end of the cutout part 421 in a direction opposite the support part 410 from the cutout part 421. According to an embodiment, different catching projections 422 formed in different cutout portions 421 may be symmetrically formed with respect to the opening 430. According to an embodiment, at least one catching projection 422 may be formed on each cutout part 421, and a plurality of catching projections 422 may be formed on a plurality of cutout parts 421. According to an embodiment, the plurality of catching projections 422 may be disposed to face each other in the direction of the opening 430.

Referring to FIG. 6B, FIG. 6B may be a top view illustrating the fastening member/fastener 400 when viewed from the top of the fastening member 400. With reference to FIG. 6B, the coupling part 420 in the shape of a planar may be substantially circular. According to various embodiments, the cross-sectional shape of the opening 430 formed across the coupling part 420 and the support part 410 may be substantially circular. According to an embodiment, a shape (e.g., cross-section) of the support part 410 may be substantially rectangular when viewed from the top of the support part 410. According to an embodiment, the horizontal length (e.g., a length in the x-axis direction) of the support part 410 may be formed in a range that is at least greater than a diameter of the communicating part 112a. According to an embodiment, the vertical length (e.g., a length in the y-axis direction) of the support part 410 may be equal to or greater than a maximum diameter of the coupling part 420. According to an embodiment, the horizontal length of the support part 410 may be greater than the vertical length of the

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support part **410**. According to an embodiment, an area formed by a difference between a longer length and a shorter length among the horizontal length and the vertical length of the support part **410**, that is, an area excluding an area formed by a cross-section of the coupling part **420** from an area formed by a cross-section of the support part **410**, may form a step (e.g., the stepped part **411**) with the coupling part **420**.

With reference to FIG. 6C, a cross-section of the fastening member/fastener **400** may be a cross-section in which the fastening member/fastener **400** is cut along line aa' in FIG. 6B. According to various embodiments, the opening **430** may be formed inside of the support part **410** and the coupling part/coupler **420**. According to an embodiment, the opening **430** may be formed such that the diameter thereof decreases as the opening **430** progresses in a direction (e.g., in the z-axis direction) from the support part **410** toward the coupling part **420**. According to various embodiments, at least a portion of the digital pen **201** may be inserted into the opening **430** by inserting the digital pen **201** from the connecting passage **112a** into the storage space **112**. According to an embodiment, the coupling part **420** may be coupled to the digital pen **201** by inserting at least a portion of the digital pen **201** (e.g., the second end **300b** in FIG. 4) into the opening **430**. According to an embodiment, a maximum diameter of the digital pen **201** may be less than or equal to a diameter of the communicating part **112a**. According to an embodiment, a maximum diameter of the opening **430** may be less than or equal to a diameter of the connecting passage **112a**.

With reference to FIG. 6C, the cutout part **421** may be partially deformed in shape by an external force to form a repulsive force. For example, the cutout part **421** formed on the right side may be partially bent outwardly when subjected to an external force outwardly (e.g., in the x-axis direction), and may apply a force inwardly (e.g., in the opposite direction of the x-axis) by a repulsive force and/or an elastic force against the deformation.

With reference to FIG. 6D, the cutout part **421** may be a plurality of pieces. The coupling part **420** may be substantially circular in shape (e.g., in cross-section) when viewed from the top of the coupling part, or may be a structure formed that the coupling part **420** formed in the circular shape is cut out. Therefore, the number of cutout parts **421** may be the same as the number of the cutout clearances **423**. According to various embodiments, the number of cutout parts **421** may be an even number. According to an embodiment, the number of cutout parts **421** may be four. According to an embodiment, the number of cutout parts **421** may be two. According to an embodiment, each of the plurality of cutout parts **421** may include the catching projection **422**. According to an embodiment, only a portion of the plurality of cutout parts **421** may include the catching projection **422**, and the remaining portion of the cutout parts **421** may not include the catching projection **422**.

According to various embodiments, the fastening member/fastener **400** may include a material with elasticity. According to an embodiment, the cutout part **421** of the fastening member **400** may include a material that may be temporarily deformed in shape by an external force. Since the fastening member **400** is coupled with the digital pen **201** by temporarily being deformed upon insertion of the digital pen **201** and then restored, a resilient, but rigid material may be preferred. According to an embodiment, the fastening member **400** may include a self-lubricating material to facilitate insertion of the digital pen **201**. According to an embodiment, the fastening member/fastener **400** may

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include a resilient material (e.g., rubber and silicone). According to an embodiment, the fastening member/fastener **400** may include a rigid material with self-lubricating property (e.g., polyoxymethylene (POM), polyamide (PA), polypropylene (PP), polycarbonate (PC), polyethylene (PE), and/or thermoplastic polyurethane (TPU)).

FIG. 7A is a perspective view illustrating the second end **300B** of the digital pen according to various embodiments.

FIG. 7B is a top view illustrating the digital pen **201** according to various embodiments.

FIG. 7C is a cross-sectional view illustrating the second end **300B** of the digital pen according to various embodiments.

FIG. 7C may be a cross-sectional view illustrating the second end **300B** of the digital pen **201** that cut along line bb' in FIG. 7B.

With reference to FIGS. 7A, 7B, and 7C, the second end **300B** may include a top portion **701** and a bottom portion **702**. According to an embodiment, the top portion **701** may include an inclined portion **710** and a catching groove **720**. According to an embodiment, the bottom portion **702** may include a planar portion **730**. According to an embodiment, the pen tip **321** may penetrate and protrude from the center of the second end **300B**.

According to various embodiments, the inclined portion **710** may include a slope formed by a difference in diameters of areas corresponding to respective portions of the vertical direction (e.g., the z-axis direction) of the second end **300b**. According to an embodiment, at least a portion of the surface of the second end **300B** may include an area that gradually increases in diameter as the second end **300B** progresses in the opposite direction of the z-axis. Therefore, as the diameter increases, a slope may be formed on the surface of the second end **300b**. According to an embodiment, the inclined portion **710** may include a structure that increases in diameter from the second end **300B** toward the first end (e.g., the first end **300A** in FIG. 4) (e.g., in the opposite direction of the z-axis).

According to various embodiments, the catching groove **720** may include a groove formed on the surface of the second end **300b**. The catching groove **720** may include, for example, an area that decreases in diameter of the second end **300b** adjacent to an area that has a maximum diameter of a portion of the inclined portion **710**. According to an embodiment, a diameter of the area in which the catching groove **720** is formed may be less than a maximum diameter among diameters of the area in which the inclined portion **710** is formed. According to an embodiment, the catching groove **720** may be formed in an area in the opposite direction of the z-axis from the inclined portion **710**. According to an embodiment, the catching groove **720** may form a stepped structure because of a difference in diameter with the inclined portion **710**.

According to various embodiments, the planar portion **730** may include a planar area. The planar area may include a planar surface perpendicular to the z-axis. According to an embodiment, the planar portion **730** may be a portion of the surface of the pen housing **300** that includes a planar surface that is perpendicular to the z-axis. With reference to FIG. 7B, a cross sectional area of the bottom portion **702** may be formed over a larger area than a cross sectional area of the top portion **701**. According to an embodiment, the cross-sectional area of the bottom portion **702** may be larger than the cross-sectional area of the top portion **701**. According to an embodiment, in the cross-sectional area of the bottom

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portion **702**, at least some of the area, other than the cross-sectional area of the top portion **701**, may include the planar portion **730**.

With reference to FIG. 7B, the cross-section of the top portion **701** may include a substantially circular shape. According to an embodiment, each of the areas included in the top portion **701** (e.g., the inclined portion **710** and the catching groove **720**) and the pen tip **321** may include a substantially circular shape in cross-section. According to an embodiment, the cross-section of each area included in the top portion **701** may form concentric circles. According to various embodiments, the cross-section of the bottom portion **702** may be larger than the cross-section of the top portion **701**. According to an embodiment, the cross-section of the bottom portion **702** may include a rectangular-like shape or a substantially elliptical shape. According to an embodiment, the cross-section of the bottom portion **702** may be an elliptical shape formed by a long axis formed in the x-axis direction and a short axis formed in the y-axis direction. According to an embodiment, a short axis of the cross-section of the bottom portion **702** may be equal to or greater than a maximum diameter of the cross-section of the top portion **701**. According to an embodiment, the planar portion **730** may be formed among the area of the bottom portion **702**, in a range that exceeds an area with a maximum diameter of the top portion **701**.

FIG. 8A is a cross-sectional view illustrating a process of coupling the digital pen with the fastening member according to various embodiments.

FIG. 8B is a cross-sectional view illustrating that the digital pen and fastening member are coupled according to various embodiments.

With reference to FIGS. 8A and 8B, the fastening member/fastener **400** may be coupled with the digital pen **201**. According to various embodiments, the digital pen **201** may be inserted into an electronic device (e.g., the electronic device **101** in FIG. 1). According to an embodiment, the digital pen **201** may be inserted from a hole (e.g., the hole **111** in FIG. 2) formed in at least some area of a housing (e.g., the housing **110** in FIG. 2) of the electronic device **101** and stored in a storage space (e.g., the storage space **112** in FIG. 2) extending from the hole **111**. According to an embodiment, the digital pen **201** may be coupled with the fastening member **400** disposed in a mounting area (e.g., the mounting area **112c** in FIG. 5C) through a communicating part (e.g., the connecting passage **112a** in FIG. 5C) included in the storage space **112**.

According to various embodiments, the digital pen **201** may be coupled with the coupling part **410** included in the fastening member **400**. According to an embodiment, the second end **300b** included in the digital pen **201** may be coupled with the coupling part **410**. According to various embodiments, the digital pen **201** may be inserted into the opening **430** formed in the center of the coupling part **410**. According to an embodiment, the digital pen **201** may be stored along a direction (e.g., the z-axis direction) from the hole **111** through the connecting passage **112a** and inserted into the opening **430** in the same direction (e.g., the z-axis direction).

With reference to FIG. 8A, the coupling part **410** may apply a buffering force to the digital pen **201** as the digital pen **201** is inserted. The buffering force may be a force that, in response to an external force that inserts the digital pen **201**, pushes the digital pen **201** in a direction opposite to the external force. According to an embodiment, a direction in which the digital pen **201** is inserted may be in the z-axis direction, and the buffering force may be formed in a

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direction opposite to the z-axis. According to various embodiments, a minimum diameter of the coupling portion **410**, for example, a minimum diameter of a diameter formed by the opening **430** corresponding to the coupling portion, may be less than a maximum diameter of the inclined portion **710** of the digital pen **201**. According to an embodiment, a minimum diameter of the inclined portion **710** may be less than or equal to a minimum diameter of the opening **430**. According to an embodiment, at least a portion of the inclined portion **710** may push the cutout part **421** outwardly by the slope formed by the inclined portion **710** as the inclined portion **710** is inserted into the opening **430** and then advanced in the z-axis direction. The outward direction may be an outward direction that includes, for example, the x-axis direction and/or a direction opposite to the x-axis, relative to the opening **430**. According to an embodiment, the smallest diameter of an area of the opening **430** may be an area corresponding to an area in which the catching projection **422** is formed. According to an embodiment, the catching projection **422** may include, in a portion thereof, an inclined surface **421s** corresponding to the inclined portion **710**. According to an embodiment, the inclined surface **421s** may have the same degree of slope corresponding to the inclined portion **710**, and the inclined portion **710** may be gently pushed, in contact with the inclined surface **421s**, in a direction of the opening **430** (e.g., in the z-axis direction). According to an embodiment, the cutout portion **421** may open as the digital pen **201** is inserted by the slope of the inclined portion **710**. According to an embodiment, the cutout portion **421** may open outwardly (e.g., in a direction opposite the opening **430**) upon insertion of the digital pen **201**. According to an embodiment, the open cutout portion **421** may apply a repulsive force inwardly (e.g., in a direction toward the opening **430**). According to an embodiment, the cutout portion **420** may deliver a repulsive force to the inclined portion **710** through a contact surface with the inclined portion **710**. According to an embodiment, the repulsive force delivered by a slope angle of the inclined portion **710** may include a force in a direction opposite to the insertion (e.g., in a direction opposite to the z-axis). For example, a combined force of repulsive force may be formed in the vertical direction of the inclined portion **710** by the slope angle, and a force in a direction opposite the z-axis may be applied to the digital pen **201** upon decomposition of the combined force. According to an embodiment, a force that pushes the digital pen **201** in a direction opposite to the z-axis may be formed by a force that is formed by a combined force of the repulsive force of the cutout part **421** and the normal drag force of the inclined portion **710**. According to an embodiment, the buffering force may include a force that pushes the digital pen **201** in a direction opposite to the z-axis.

With reference to FIG. 8B, the fastening member/fastener **400** may be coupled to the digital pen **201**. According to various embodiments, in case of insertion of the digital pen **201**, the catching projection **422** of the fastening member **400** may pass over the inclined portion **710** and rest in the catching groove **720**.

According to various embodiments, the insertion of the digital pen **201** is completed when the planar portion **730** and the cutout part **421** come into contact. According to an embodiment, the cutout part/cutout **421** may be interrupted by the planar portion **730**. According to an embodiment, a contact portion **421c** formed at an end of the cutout portion **421** may include a planar shape corresponding to the planar portion **730**. According to an embodiment, the planar portion **730** is interrupted by the contact portion **421c** and may no

longer advance in the insertion direction (e.g., in the z-axis direction), and the insertion may be completed.

According to an embodiment, a diameter formed by a cross-section of an area corresponding to the catching groove 720 may be less than or equal to a diameter of the opening 430 corresponding to the catching projection 422. According to an embodiment, the diameter of the opening 430 formed inside the catching projection 422 may be less than a maximum diameter of the inclined portion 710. According to an embodiment, the catching projection 422 may be interrupted by a step formed by a difference between a diameter of the catching groove 420 and a maximum diameter of the inclined portion 710. In other words, in case that the catching projection 422 is coupled to the catching groove 720, the digital pen 201 may be coupled to the fastening member/fastener 400 with the inclined portion 710 interrupted by the catching projection 422. According to an embodiment, the coupling of the catching projection 422 and the catching groove 720 may allow the digital pen 201 to be coupled without being separated in a direction opposite to the insertion (e.g., in a direction opposite to the z-axis).

An electronic device according to various example embodiments may include: a housing; a hole formed in a portion of the housing; a storage space that is connected, directly or indirectly, to the hole and extends into the housing; and a fastening member/fastener that is disposed in the storage space and fastens an external electronic device, in which the fastening member may include: a support part for fixing the fastening member to the storage space; a coupling part which extends from the support part and in the center of which an opening for inserting the external electronic device is formed; and two or more cutout parts formed by cutting out a portion of the coupling part, in which the cutout parts may form a repulsive force against an external force in an inward direction or an outward direction of the opening.

In addition, the electronic device may include the two or more catching projections protruding from a portion of each of the two or more cutout parts, and disposed to face each other in a direction toward the opening.

In addition, the two or more cutout parts may form a symmetrical structure with respect to the opening.

In addition, a cross-section of the opening may be circular.

In addition, the two or more cutout parts may form the repulsive force in an inward direction of the opening when opened apart from each other in an outward direction of the opening.

In addition, the storage space may include a communicating part into which the external electronic device is inserted, and an interruption portion configured to interrupt the support part to move away corresponding to the support part, in which the support part may be fixed to the storage space by the interruption portion.

In addition, a diameter of the support part may be greater than a diameter of the communicating part.

In addition, a maximum diameter of the opening may be less than or equal to the diameter of the communicating part.

In addition, a diameter of the opening formed by the two or more catching projections facing each other in an area of the opening may be less than a maximum diameter of the opening.

In addition, the fastening member may include four cutout parts/cutouts.

In addition, the fastening member (or fastener) may include, for example and without limitation, any one or a combination of polyoxymethylene (POM), polyamide (PA),

polypropylene (PP), polycarbonate (PC), polyethylene (PE), and thermoplastic polyurethane (TPU), or any suitable adhesive or other fastener.

In addition, the external electronic device is an electronic pen, in which the electronic pen may include: a pen housing; a first end formed at both ends of the pen housing, at least a portion of which protrudes outside of the hole upon the insertion; and a second end positioned inside the storage space upon the insertion, and in which the second end may include: an inclined portion forming a slope on a surface of the pen housing by increasing in diameter in a direction from the second end toward the first end; a catching groove formed at an end of the inclined portion in a direction toward the first end, and including a diameter smaller than a maximum diameter formed by the inclined portion; and a planar portion extending from the catching groove and configured to interrupt the catching groove in a direction perpendicular to the diameter.

In addition, a minimum diameter of the cutout part may be less than a maximum diameter of the inclined portion of the electronic pen and greater than or equal to a minimum diameter of the inclined portion of the electronic pen. A maximum diameter of the inclined portion may be greater than the minimum diameter of the cutout part.

In addition, the cutout part may be opened in an outward direction of the opening upon the insertion by the inclined portion of the electronic pen, in which the cutout part which is opened may generate a repulsive force that is applied to an area in which an end of the cutout part is in contact with the inclined portion in an inward direction of the opening, and a buffering force that pushes the electronic pen by a slope of the inclined portion in an outward direction of the housing.

In addition, the catching groove of the electronic pen, upon the insertion, may interrupt the coupling portion of the fastening member in a direction opposite to the insertion.

In addition, the planar portion of the electronic pen, upon insertion, may interrupt the coupling portion with respect to a direction of the insertion.

An external electronic device according to various example embodiments may include: a housing; a first end formed at one end of the housing; and a second end formed at the other end of the housing in a direction opposite to the first end, and in which the second end may include: an inclined portion forming a slope on a surface of the pen housing by increasing in diameter in a direction from the second end toward the first end; a catching groove formed at an end of the inclined portion in a direction toward the first end, and including a diameter smaller than a maximum diameter formed by the inclined portion; and a planar portion extending from the catching groove and configured to interrupt the catching groove in a direction perpendicular to the diameter.

In addition, the electronic device may further include: a tip disposed inside the housing, extending to the outside through the second end, and including a dielectric coil of a plurality of windings therein; and a printed circuit board electrically connected to the dielectric coil.

The electronic device according to various example embodiments may be a device of various shapes. The electronic device may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. The electronic devices according to embodiments of the present document are not limited to the devices described above.

The various embodiments of the present document and the terms used therein are not intended to limit the technical features described in the present document to specific embodiments, but should be understood to include various modifications, equivalents, or substitutions of the corresponding embodiments. In connection with the description of the drawings, the similar reference numerals may be used for the similar or relevant constituent elements. The singular form of a noun corresponding to an item may include one or more of the items, unless the relevant context clearly indicates otherwise. In the present document, each of the phrases “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B or C,” “at least one of A, B and C,” and “at least one of A, B, or C” may include any of the items listed together in the corresponding phrase among those phrases, or any possible combination thereof. The terms such as “first,” “second,” or “first” or “second” may be used simply to distinguish a constituent element from other corresponding constituent elements and do not limit the constituent elements in any other respect (e.g., importance or order). In case that any (e.g., a first) constituent element is referred to as “coupled” or “connected” to another (e.g., a second) constituent element, with or without the terms “functionally” or “communicationally”, it means that the constituent element may be connected to the other constituent element directly (e.g., wired), wirelessly, or through at least a third constituent element(s).

As used in various embodiments of the present document, the term “module” may include a unit implemented in hardware, software, or firmware, and may be used interchangeably with terms such as logic, logic block, component, or circuit, for example. The module may be a component that is integrally constituted, or a minimum unit of the component, or a part thereof, that performs one or more functions. For example, according to an embodiment, the module may be implemented in the form of an application-specific integrated circuit (ASIC). Thus, each “module” herein may comprise circuitry.

Various embodiments may be implemented as software (e.g., the program 140) including one or more instructions stored on a storage medium (e.g., the internal memory 136 or external memory 138) readable by a machine (e.g., the electronic device 101). For example, a processor (e.g., the processor 120) of the machine (e.g., the electronic device 101) may call and execute at least one of one or more instructions stored on the storage medium. This enables the machine to be operated to perform at least one function according to the at least one instruction called. One or more of the instructions that are described above may include code generated by a compiler or code that may be executed by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Here, “non-transitory” only means that the storage medium is a tangible device and does not include signals (e.g., electromagnetic waves), and this term does not distinguish between the case where the data is stored on the storage medium permanently and the case where the data is stored temporarily.

In some embodiments, methods according to various embodiments disclosed herein may be provided in a computer program product. The computer program product is a commodity and may be traded between a seller and a buyer. The computer program product may be distributed in the form of a device-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or it may be distributed online (e.g., downloaded or uploaded) through an application store (e.g., Play Store™) or directly between two user

devices (e.g., smartphones). In case of online distribution, at least a portion of the computer program product may be stored, or at least temporarily generated, on a device-readable storage medium, such as a manufacturer’s server, an application store’s server, or a memory of a relay server.

According to various embodiments, each constituent element (e.g., module or program) of the above-described constituent elements may include a single or plurality of objects, and some of the plurality of objects may be disposed separately in different constituent elements. In various embodiments, one or more constituent elements of the corresponding constituent elements described above, or operations may be omitted, or one or more other constituent elements or operations may be added. Alternatively or additionally, a plurality of constituent elements (e.g., modules or programs) may be integrated into a single constituent element. In this case, the integrated constituent element may perform one or more functions of the constituent element of each of the plurality of constituent elements in the same or similar manner as performed by the corresponding constituent element of the plurality of constituent elements prior to the integration. According to various embodiments, the operations performed by a module, program, or other constituent element may be executed sequentially, in parallel, iteratively, or heuristically, or one or more of the above operations may be executed in a different order, omitted, or one or more other operations may be added.

While the disclosure has been illustrated and described with reference to various embodiments, it will be understood that the various embodiments are intended to be illustrative, not limiting. It will further be understood by those skilled in the art that various changes in form and detail may be made without departing from the true spirit and full scope of the disclosure, including the appended claims and their equivalents. It will also be understood that any of the embodiment(s) described herein may be used in conjunction with any other embodiment(s) described herein.

The invention claimed is:

1. An electronic device comprising:

- a housing;
 - a hole formed in a portion of the housing;
 - a storage space connected to the hole and provided inside the housing; and
 - a fastener disposed inside the storage space and configured to fasten an external electronic device,
- wherein the fastener comprises:
- a support part configured to fix the fastener to the storage space;
 - a coupler extending from the support part and having an opening formed in the center thereof in which an external pen is insertable; and
 - two or more cutout parts formed by cutting out a portion of the coupler,
- wherein the two or more cutout parts include catching projections protruding from a portion of each of the two or more cutout parts, the catching projections being disposed to face each other in a direction toward the opening of the coupler, and
- wherein the two or more cutout parts are configured to provide a repulsive force against an external force in an inward direction or an outward direction of the opening,
- wherein the external pen comprises:
- a pen housing,
 - a first end and a second end formed at opposite ends of the pen housing, at least a portion of the first end protruding outside of the hole of the housing upon

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the insertion, wherein the second end is positioned inside the storage space upon the insertion, and wherein the second end comprises:

- an inclined portion forming a slope on a surface of the pen housing by increasing in diameter in a direction from the second end toward the first end,
- a catching groove formed at an end of the inclined portion in a direction toward the first end, and comprising a diameter smaller than a maximum diameter formed by the inclined portion, and
- a planar portion substantially perpendicular to the direction from the second end toward the first end, and

wherein the two or more cutout parts include a contact portion formed at an end of the cutout part, which comes into contact with the planar portion of the external pen.

2. The electronic device of claim 1, comprising two or more catching projections protruding from a portion of each of the two or more cutout parts, and disposed to face each other in a direction toward the opening.

3. The electronic device of claim 1, wherein the two or more cutout parts form a symmetrical structure with respect to the opening.

4. The electronic device of claim 1, wherein the two or more cutout parts are configured to form the repulsive force in an inward direction of the opening when opened apart from each other in an outward direction of the opening.

5. The electronic device of claim 1, wherein the storage space comprises a communicating part into which the external pen is to be inserted, and an interruption portion configured to interrupt the support part to move away corresponding to the support part, and

wherein the support part is fixed to the storage space by at least the interruption portion.

6. The electronic device of claim 5, wherein a diameter of the support part is greater than a diameter of a connecting passage.

7. The electronic device of claim 5, wherein a maximum diameter of the opening is less than or equal to a diameter of a connecting passage.

8. The electronic device of claim 2, wherein a diameter of the opening formed by the two or more catching projections facing each other in an area of the opening is less than a maximum diameter of the opening.

9. The electronic device of claim 1, wherein the fastener comprises four cutout parts.

10. The electronic device of claim 1, wherein the fastener comprises any one or a combination of polyoxymethylene

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(POM), polyamide (PA), polypropylene (PP), polycarbonate (PC), polyethylene (PE), and thermoplastic polyurethane (TPU).

11. The electronic device of claim 1, wherein the external electronic device is an electronic pen,

wherein the electronic pen comprises:

- a pen housing;
- a first end, at least a portion of which is to protrude outside of the hole upon the insertion; and
- a second end configured to be positioned inside the storage space upon the insertion, and

wherein the second end comprises:

- an inclined portion forming a slope on a surface of the pen housing by increasing in diameter in a direction from the second end toward the first end;
- a catching groove formed at an end of the inclined portion in a direction toward the first end, and comprising a diameter smaller than a maximum diameter formed by the inclined portion; and
- a planar portion extending from the catching groove and configured to interrupt the catching groove in a direction perpendicular to the diameter.

12. The electronic device of claim 11 wherein a minimum diameter of the cutout parts is less than a maximum diameter of the inclined portion of the electronic pen and greater than or equal to a minimum diameter of the inclined portion of the electronic pen, and

wherein a maximum diameter of the inclined portion is greater than the minimum diameter of the cutout parts.

13. The electronic device of claim 12, wherein the cutout parts are configured to be opened in an outward direction of the opening upon the insertion by the inclined portion of the electronic pen, and

wherein the cutout parts which are configured to be opened to generate a repulsive force to be applied to an area in which an end of the cutout parts is in contact with the inclined portion in an inward direction of the opening, and a buffering force to push the electronic pen by a slope of the inclined portion in an outward direction of the housing.

14. The electronic pen of claim 12, wherein the catching groove of the electronic pen, upon the insertion, is configured to interrupt a coupling portion of the fastener in a direction opposite to the insertion.

15. The electronic pen of claim 12, wherein the planar portion of the electronic pen, upon insertion, is configured to interrupt a coupling portion with respect to a direction of the insertion.

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